

CS 109ACE Week 3 Practice Problems

1 Independence and Inclusion–Exclusion

Problem 1: Practice 1: Three Clubs

(slide 11)

At a school, 50% of students are in a CS club, 40% in a math club, and 30% in a stats club. 20% are in CS and math, 15% in CS and stats, 10% in math and stats, and 5% are in all three.

- (a) What fraction of students are in at least one of the clubs?
- (b) What fraction are in none?
- (c) What fraction are in the CS club only?

Solution. Let C , M , T be the events that a random student is in the CS, math, and stats club.

- (a) By inclusion–exclusion for three events,

$$P(C \cup M \cup T) = 0.50 + 0.40 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05 = 0.80.$$

- (b) “None” is the complement of “at least one,” so $P(\text{none}) = 1 - 0.80 = 0.20$.
- (c) To get “CS only,” start from $P(C)$ and remove both overlaps. Subtracting $P(CM)$ and $P(CT)$ removes the triple overlap twice, so add it back once:

$$P(C \cap M^c \cap T^c) = P(C) - P(CM) - P(CT) + P(CMT) = 0.50 - 0.20 - 0.15 + 0.05 = 0.20.$$

Check: the seven Venn regions are 0.20, 0.15, 0.10, 0.15, 0.10, 0.05, 0.05, which sum to 0.80.



2 Counting

Problem 2: Warm-up: Entry Codes

(slide 14)

Entry codes are 6 digits and may start with 0 (e.g., 379009, 012345, 000000).

- (a) How many codes are possible?
- (b) How many are the same two digits repeated three times (121212, 333333)?
- (c) How many use each of the digits 1–6 exactly once (254631)?
- (d) *Stretch:* a code is valid if its digits sum to exactly 9. What is $P(\text{valid})$ for a code picked uniformly at random?

Solution.

- (a) There are 10 choices for each of the 6 positions, so by the product rule there are $10^6 = 1,000,000$ codes.
- (b) The first two digits determine the whole code, so there are $10 \cdot 10 = 100$ such codes. (Careful: 333333 is counted once, as the pair “33.”)
- (c) We are arranging 6 distinct digits in a row: $6! = 720$.
- (d) We need the number of digit strings with $d_1 + d_2 + \cdots + d_6 = 9$ and each $d_i \geq 0$. No digit can exceed 9, because the total is only 9, so the upper limit never matters. By stars and bars (9 stars and 5 bars),

$$\#\{\text{valid codes}\} = \binom{14}{5} = 2002, \quad P(\text{valid}) = \frac{2002}{10^6} \approx 0.002.$$

■

Problem 3: Practice 2: Committees

(slide 22)

- (a) An animal club has 9 antelopes and 11 beavers. How many 7-member committees have exactly 4 beavers and 3 antelopes?
- (b) From 2 cats and 5 dogs, a committee of 3 is formed. How many committees include at least one cat?

Solution.

- (a) Choose the beavers and the antelopes as two separate steps (product rule):

$$\binom{11}{4} \binom{9}{3} = 330 \cdot 84 = 27,720.$$

- (b) “At least one” suggests the complement: all committees minus the committees with no cat,

$$\binom{7}{3} - \binom{5}{3} = 35 - 10 = 25.$$

Check by cases: exactly one cat gives $\binom{2}{1} \binom{5}{2} = 20$, and two cats give $\binom{2}{2} \binom{5}{1} = 5$, for 25 in total.
Trap: “pick one cat, then any 2 of the other 6 animals” gives $2 \cdot 15 = 30$, which double-counts the committees with both cats.

■

Problem 4: Practice 3: BASKETBALL

(slide 24)

- (a) How many distinct 10-letter arrangements of the letters in BASKETBALL are there?
- (b) How many of those arrangements have the two L’s next to each other?

Solution.

- (a) BASKETBALL has 10 letters: B, A, and L appear twice each, and S, K, E, T once each. Dividing out the orderings of identical letters,

$$\frac{10!}{2! 2! 2!} = \frac{3,628,800}{8} = 453,600.$$

- (b) Glue the two L's into one block "LL." That leaves 9 objects (B twice, A twice, LL, S, K, E, T), so

$$\frac{9!}{2!2!} = \frac{362,880}{4} = 90,720.$$

There is no extra factor for ordering inside the block, because the two L's are identical. ■

Problem 5: Practice 4: Bookshelf

(slide 26)

Three different hardcover books and five different paperbacks go on a shelf. How many arrangements keep all the hardcovers together *and* all the paperbacks together?

Solution. Build an arrangement in three steps (product rule): order the two blocks (hardcovers first or paperbacks first) in $2! = 2$ ways, order the hardcovers inside their block in $3! = 6$ ways, and order the paperbacks inside their block in $5! = 120$ ways. The total is

$$2 \cdot 6 \cdot 120 = 1440.$$

For comparison, there are $8! = 40,320$ arrangements in all. ■

Problem 6: Practice 5: Splitting \$7

(slide 29)

You have 7 identical \$1 bills to split among Amber, Ben, and Curtis.

- (a) How many ways can you split them (someone may get \$0)?
- (b) How many ways if each person gets at least \$1?

Solution. Putting n identical items into r distinct bins can be done in $\binom{n+r-1}{r-1}$ ways (stars and bars).

- (a) Use 7 stars and 2 bars (3 people need 2 dividers): $\binom{9}{2} = 36$. For example, $\star\star | \star | \star\star\star$ means Amber gets \$2, Ben \$1, and Curtis \$4.
- (b) Give everyone \$1 first, then split the remaining \$4 freely: $\binom{4+2}{2} = \binom{6}{2} = 15$.

■

Problem 7: Counting Challenge: Six-Card Poker

(slide 31)

A casino deals 6-card poker hands from a standard 52-card deck (hands are unordered).

- (a) *Three pair*: 2 cards of one rank, 2 of another, and 2 of a third. What is $P(\text{three pair})$?
- (b) *Pyramid*: 3 cards of one rank, 2 of another, and 1 of a third. What is $P(\text{pyramid})$?

Solution. The sample space is all unordered 6-card hands, $|S| = \binom{52}{6} = 20,358,520$, all equally likely.

- (a) Choose the 3 ranks (unordered, since all three play the same role), then 2 of the 4 suits for each:

$$|E| = \binom{13}{3} \binom{4}{2}^3 = 286 \cdot 216 = 61,776,$$

$$P(\text{three pair}) = \frac{61,776}{20,358,520} = \frac{594}{195,755} \approx 0.3034\%.$$

- (b) Now the three ranks play *different* roles (triple, pair, single), so choose them in order: $13 \cdot 12 \cdot 11 = 1716$ ways. Then choose suits: $\binom{4}{3}\binom{4}{2}\binom{4}{1} = 96$. So

$$|E| = 1716 \cdot 96 = 164,736,$$

$$P(\text{pyramid}) = \frac{164,736}{20,358,520} = \frac{1584}{195,755} \approx 0.8092\%.$$

■

Problem 8: Poker Straight

(slide 33)

Draw a 5-card hand from a standard 52-card deck. A straight is 5 cards of consecutive ranks, e.g., 2-3-4-5-6. An ace can be low (A-2-3-4-5) or high (10-J-Q-K-A). Suits don't matter (count straight flushes too). What is $P(\text{straight})$? First, describe the sample space and the event in words.

Solution. S is the set of all unordered 5-card hands, so $|S| = \binom{52}{5} = 2,598,960$. E is the set of all straights. Build one step by step (product rule): pick the lowest rank (A, 2, ..., 10), which gives 10 ways and fixes all 5 ranks; then pick a suit for each card, which gives $4^5 = 1024$ ways. So

$$|E| = 10 \cdot 1024 = 10,240, \quad P(\text{straight}) = \frac{10,240}{2,598,960} \approx 0.0039.$$

Why not multiply by 2 for “up or down”? Naming a straight by its lowest card already counts each hand exactly once. ■

3 Random Variables and PMFs

Problem 9: Practice 6: Max of Two Dice

(slide 38)

Roll two fair dice. Let M be the larger of the two values (if they tie, M is that value).

- Find the PMF of M : $P(M = k)$ for $k = 1, \dots, 6$.
- Check that your PMF sums to 1.
- Find $P(M \leq 3)$ and $P(M \geq 5)$.

Solution.

- $M \leq k$ exactly when both dice are at most k , which happens in k^2 of the 36 equally likely outcomes. So

$$P(M = k) = P(M \leq k) - P(M \leq k-1) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36},$$

which gives $\frac{1}{36}, \frac{3}{36}, \frac{5}{36}, \frac{7}{36}, \frac{9}{36}, \frac{11}{36}$ for $k = 1, \dots, 6$.

- $1 + 3 + 5 + 7 + 9 + 11 = 36$, so the probabilities sum to $\frac{36}{36} = 1$.
- $P(M \leq 3) = \frac{9}{36} = \frac{1}{4}$, and $P(M \geq 5) = \frac{9+11}{36} = \frac{20}{36} = \frac{5}{9}$.

■

4 The Binomial

Problem 10: Practice 7: Clinical Trial

(slide 44)

A new medicine is tested on 40 patients. Each patient is cured independently with probability $3/10$. Let X be the number of cured patients.

- (a) What distribution does X have?
- (b) What is $P(X = 38)$?
- (c) What is the most likely number of cures, and what is its probability?

Solution.

- (a) There are 40 independent trials with the same success probability 0.3, so $X \sim \text{Bin}(40, 0.3)$.
- (b) By the Binomial PMF,

$$P(X = 38) = \binom{40}{38} (0.3)^{38} (0.7)^2 = 780 \cdot (0.3)^{38} \cdot 0.49 \approx 5.2 \times 10^{-18}.$$

This is essentially impossible: we expect around $40 \cdot 0.3 = 12$ cures.

- (c) The PMF peaks at $k = 12$:

$$P(X = 12) = \binom{40}{12} (0.3)^{12} (0.7)^{28} \approx 0.137.$$

Even the most likely value has under a 14% chance, because the probability is spread over many values.

■

Problem 11: Practice 8: Planting Tomatoes

(slide 46)

Each tomato seed sprouts independently with probability 0.7. You plant 5 seeds.

- (a) What is the probability that more than 3 tomatoes sprout?
- (b) A classmate answers (a) with $\binom{5}{4}(0.7)^4$. What's wrong with that?

Solution.

- (a) Let $X \sim \text{Bin}(5, 0.7)$. “More than 3” means $X = 4$ or $X = 5$, which are mutually exclusive, so we add:

$$P(X = 4) = \binom{5}{4} (0.7)^4 (0.3) = 0.36015, \quad P(X = 5) = (0.7)^5 = 0.16807,$$

$$P(X > 3) = 0.36015 + 0.16807 = 0.52822.$$

- (b) $\binom{5}{4}(0.7)^4 = 1.2005 > 1$, which is impossible for a probability. The expression leaves out the (0.3) factor for the fifth seed, so the “choose 4 slots to succeed” events overlap, and the outcome where all 5 sprout is counted 5 times. “At least k ” must be computed as a sum of PMF terms.

■

Problem 12: Practice 9 (Challenge): Random Teammates

(slide 48)

12 people are split into teams with a random number generator: each person independently gets a team number from 1 to 6, uniformly at random. What is the probability that at least one person has no teammate (ends up alone on a team)?

Solution. Each of the 12 people picks 1 of 6 teams, so there are $|S| = 6^{12}$ equally likely outcomes. Let A_i be the event that team i has exactly one member. We want $P(A_1 \cup \dots \cup A_6)$, so we use inclusion-exclusion.

Count the outcomes where j specific teams are singletons: choose the people for those teams, in order, in $12 \cdot 11 \cdot \dots \cdot (12 - j + 1) = \frac{12!}{(12-j)!}$ ways, and send everyone else to the other $6 - j$ teams in $(6 - j)^{12-j}$ ways. Therefore

$$P\left(\bigcup_{i=1}^6 A_i\right) = \sum_{j=1}^6 (-1)^{j+1} \binom{6}{j} \frac{12!}{(12-j)!} \cdot \frac{(6-j)^{12-j}}{6^{12}}.$$

Equivalently, 262,755,576 of the $6^{12} = 2,176,782,336$ outcomes have no singleton team, so

$$P = 1 - \frac{262,755,576}{2,176,782,336} = \frac{2,953,745}{3,359,232} \approx 0.879.$$

Surprise: almost 88%! On average about 1.6 teams end up with a lone member. (A simulation with 200,000 trials gives about 0.880.) ■

5 Moments

Problem 13: Practice 10: Marble Points

(slide 62)

Draw one marble: red with probability 0.5, green 0.3, blue 0.2. Let $X = 1$ for red, 2 for green, 3 for blue.

- (a) Find $E[X]$.
- (b) Find $E[X^2]$ (use LOTUS).
- (c) Find $\text{Var}(X)$ and $\text{SD}(X)$.
- (d) You score $S = 10X - 4$ points. Find $E[S]$ and $\text{SD}(S)$.

Solution.

- (a) $E[X] = 1(0.5) + 2(0.3) + 3(0.2) = 0.5 + 0.6 + 0.6 = 1.7$.
- (b) By LOTUS, square the *values* and keep the probabilities: $E[X^2] = 1(0.5) + 4(0.3) + 9(0.2) = 0.5 + 1.2 + 1.8 = 3.5$.
- (c) $\text{Var}(X) = E[X^2] - (E[X])^2 = 3.5 - 1.7^2 = 3.5 - 2.89 = 0.61$, and $\text{SD}(X) = \sqrt{0.61} \approx 0.78$.
- (d) By linearity, $E[S] = 10 E[X] - 4 = 17 - 4 = 13$. Shifting by -4 doesn't change the spread, so

$$\text{Var}(S) = 10^2 \cdot 0.61 = 61, \quad \text{SD}(S) = 10 \cdot \text{SD}(X) = \sqrt{61} \approx 7.81.$$

■

Problem 14: Practice 11: Expected Max of Two Dice

(slide 64)

Back to Practice 6: roll two fair dice and let M be the larger value, so $P(M = k) = \frac{2k-1}{36}$.

(a) Find $E[M]$.

(b) Let L be the smaller value. Find $E[L]$ without working out a new PMF.

Hint for (b): what is $M + L$?

Solution.

(a) Using the PMF from Practice 6,

$$E[M] = \sum_{k=1}^6 k \cdot \frac{2k-1}{36} = \frac{1+6+15+28+45+66}{36} = \frac{161}{36} \approx 4.47.$$

(b) On every roll, $M + L = (\text{die 1}) + (\text{die 2})$. By linearity,

$$E[M] + E[L] = E[\text{die 1}] + E[\text{die 2}] = 3.5 + 3.5 = 7, \quad \text{so} \quad E[L] = 7 - \frac{161}{36} = \frac{91}{36} \approx 2.53.$$

M and L are dependent, but linearity of expectation doesn't need independence. *Check:* $P(L = k) = \frac{13-2k}{36}$ gives $E[L] = \frac{11+18+21+20+15+6}{36} = \frac{91}{36}$.

■

Problem 15: Practice 12: Guessing on a Quiz

(slide 66)

A quiz has 20 multiple-choice questions, 4 options each. You guess every answer at random. A correct answer is worth +1 point and a wrong answer costs $\frac{1}{3}$ point. Let C be the number correct.

(a) What distribution does C have? Find $E[C]$ and $\text{Var}(C)$.

(b) Your score is $S = C - (20 - C)/3$. Find $E[S]$ and $\text{SD}(S)$.

(c) Why do you think the penalty is exactly $\frac{1}{3}$?

Solution.

(a) There are 20 independent trials, each correct with probability $\frac{1}{4}$, so $C \sim \text{Bin}(20, \frac{1}{4})$. Then

$$E[C] = np = 20 \cdot \frac{1}{4} = 5, \quad \text{Var}(C) = np(1-p) = 20 \cdot \frac{1}{4} \cdot \frac{3}{4} = 3.75.$$

(b) Rewrite the score as a linear function of C : $S = C - \frac{20-C}{3} = \frac{4}{3}C - \frac{20}{3}$. So

$$E[S] = \frac{4}{3}(5) - \frac{20}{3} = 0, \quad \text{Var}(S) = \left(\frac{4}{3}\right)^2 \cdot 3.75 = \frac{20}{3} \approx 6.67, \quad \text{SD}(S) \approx 2.58.$$

(c) The penalty is chosen so that blind guessing is worth 0 points on average. Each guess gains 1 with probability $\frac{1}{4}$ and loses $\frac{1}{3}$ with probability $\frac{3}{4}$, and $\frac{1}{4} - \frac{3}{4} \cdot \frac{1}{3} = 0$.

■

Problem 16: Practice 13 (Challenge): Distinct Birthdays

(slide 68)

30 people have independent birthdays, each uniform over 365 days (ignore leap years).

- (a) What is the expected number of *distinct* birthdays among them?
- (b) What is the expected number of *pairs* of people who share a birthday?

Hint: use one indicator per day for (a) and one per pair of people for (b).

Solution.

- (a) For each day d , let $I_d = 1$ if at least one person has birthday d . By the complement (nobody picks d),

$$P(I_d = 1) = 1 - \left(\frac{364}{365}\right)^{30} \approx 0.0790.$$

The number of distinct birthdays is $\sum_d I_d$, so by linearity

$$E[\text{\#distinct}] = 365 \left(1 - \left(\frac{364}{365}\right)^{30}\right) \approx 28.84.$$

- (b) For each of the $\binom{30}{2} = 435$ pairs of people, let $J = 1$ if the pair shares a birthday. A given pair matches with probability $\frac{1}{365}$, so

$$E[\text{\#matching pairs}] = \frac{435}{365} \approx 1.19.$$

The indicators are dependent, but linearity still works, and no PMF is needed. (Compare: the probability of at least one shared birthday among 30 people is about 0.706.) ■

6 More Practice (take-home)

Problem 17: Take-home 1: Phone Numbers

(slide 71)

How many 7-digit phone numbers are possible if the first digit can't be 0 or 1?

Solution. By the product rule, there are 8 choices for the first digit (2–9) and 10 for each of the other 6: $8 \cdot 10^6 = 8,000,000$. ■

Problem 18: Take-home 2: Glitchy PSET App

(slide 73)

Each time you press submit, the PSET app “glitches” and reveals the solution with probability 0.05, independently. What is the probability of at least one glitch in 5 submissions?

Solution. Let X be the number of glitches in 5 submissions, so $X \sim \text{Bin}(5, 0.05)$. Using the complement,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.95)^5 = 1 - 0.7738 \approx 0.226.$$

■

Problem 19: Take-home 3: Chip Defect Detection

(slide 75)

n chips are manufactured, and exactly 1 is defective. k chips are chosen at random for testing. What is the probability that the defective chip is among the k tested chips?

Solution. There are $|S| = \binom{n}{k}$ equally likely sets of tested chips. A set contains the defective chip when it includes that chip plus $k - 1$ of the other $n - 1$, so $|E| = \binom{1}{1} \binom{n-1}{k-1}$. Then

$$P = \frac{\binom{n-1}{k-1}}{\binom{n}{k}} = \frac{k}{n}.$$

Intuition: each chip has the same chance, k/n , of being tested. ■

Problem 20: Take-home 4: More 1s Than 0s

(slide 77)

A random 10-bit string has each bit independently equal to 1 with probability $1/2$. What is the probability that it has more 1s than 0s?

Solution. Let $X \sim \text{Bin}(10, \frac{1}{2})$ be the number of 1s. “More 1s than 0s” means $X \geq 6$:

$$P(X \geq 6) = \frac{\binom{10}{6} + \binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{210 + 120 + 45 + 10 + 1}{1024} = \frac{386}{1024} = \frac{193}{512} \approx 0.377.$$

Symmetry check: $P(X \geq 6) = P(X \leq 4)$ and $P(X = 5) = \frac{252}{1024}$, so $2 \cdot 386 + 252 = 1024$. ■

Problem 21: Take-home 5: Combinatorial Proofs

(slide 79)

Prove each identity by counting the same thing two ways (a story, not algebra):

- (a) $\binom{n}{k} = \binom{n}{n-k}$
- (b) $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
- (c) $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$

Solution.

- (a) Out of n people, choosing the k who get cake is the same as choosing the $n - k$ who don't.
- (b) Fix one person, Pat. A group of k either includes Pat (choose the other $k - 1$ from the remaining $n - 1$) or doesn't (choose all k from the remaining $n - 1$). The cases don't overlap, so their counts add.
- (c) Count the subsets of n people. Each person is either in or out, so there are 2^n subsets. Alternatively, sort the subsets by their size k : there are $\binom{n}{k}$ of each size, which gives the left side.

■

Problem 22: Take-home 6: Round Table

(slide 81)

9 people sit around a round table (rotations of the same seating count as the same). How many seatings are there if 3 particular people must sit next to each other?

Solution. Glue the 3 friends into one block, which leaves 7 “units” around the table. For a circular arrangement, fix one unit's seat and arrange the other 6: $6! = 720$ ways. Then order the 3 friends inside the block: $3! = 6$ ways. The total is $720 \cdot 6 = 4320$. ■

Problem 23: Take-home 7: Lines and Triangles

(slide 83)

8 points lie in the plane, and exactly 3 of them are collinear (no other 3 are on a line).

- (a) How many distinct straight lines pass through at least 2 of the points?
- (b) How many triangles have their corners at these points?

Solution.

- (a) Every pair of points gives a line: $\binom{8}{2} = 28$. But the $\binom{3}{2} = 3$ pairs among the collinear points all give the *same* line, so there are $28 - 3 + 1 = 26$ distinct lines.

- (b) Every 3 points form a triangle, except the 3 collinear points: $\binom{8}{3} - 1 = 56 - 1 = 55$. ■

Problem 24: Take-home 8: Seven Dice

(slide 85)

Roll 7 fair six-sided dice. The probability that they sum to 10 can be written as $n/6^7$. What is n ?

Solution. Let $d_i = (\text{die } i) - 1 \geq 0$. Then $d_1 + \cdots + d_7 = 10 - 7 = 3$. By stars and bars (3 stars and 6 bars),

$$n = \binom{3+6}{6} = \binom{9}{6} = 84.$$

No d_i can exceed 5, because the total is only 3, so the upper limit on a die never matters. Then $P = \frac{84}{6^7} = \frac{84}{279,936} \approx 0.0003$. ■

Problem 25: Take-home 9: Entry Codes, Part 2

(slide 87)

Entry codes are 6 digits and may start with 0.

- (a) How many codes contain a 5-digit palindrome (e.g., 123217 or 657775)?
- (b) How many codes contain exactly three of some digit in a row (e.g., 864443, 212221, 000777), with no run longer than 3?

Solution.

- (a) A 5-digit palindrome sits either in positions 1–5 (form $ABCBA$?) or in positions 2–6 (form $?ABCBA$). Each form has 10^4 codes. A code of both forms must look like $ABABAB$, and there are 100 of those. By inclusion–exclusion, $20,000 - 100 = 19,900$.
- (b) Write D for the repeated digit and N, N' for digits different from D . The run of three can start in four places: $DDDN??$ gives $10 \cdot 9 \cdot 10 \cdot 10 = 9000$ codes; $NDDDN'?$ gives $10 \cdot 9 \cdot 9 \cdot 10 = 8100$; $?NDDDN'$ gives 8100; and $??NDDDD$ gives 9000. Codes with two runs, of the form $DDDddd$ with $d \neq D$, are counted twice, and there are $10 \cdot 9 = 90$ of them. The total is

$$9000 + 8100 + 8100 + 9000 - 90 = 34,110.$$

■

Problem 26: Take-home 10: Double Straight

(slide 89)

Six-card poker, continued: a “double straight” is 2 cards of one rank, 2 of the next rank, and 2 of the rank after that (e.g., 778899). Ranks run A, 2, ..., K, A (an ace can be low or high). What is the probability of being dealt a double straight in a 6-card hand?

Solution. There are $|S| = \binom{52}{6} = 20,358,520$ hands. The starting rank can be A, 2, ..., Q (12 choices, from A-2-3 up to Q-K-A), which fixes all three ranks. For each rank, choose 2 of the 4 suits: $\binom{4}{2}^3 = 216$. So

$$P = \frac{12 \cdot 216}{20,358,520} = \frac{2592}{20,358,520} = \frac{324}{2,544,815} \approx 0.00013.$$

(Every double straight is also a “three pair” hand.) ■

7 Other Questions in the Deck (not tagged PRACTICE)

7.1 Independence (Lecture 3 Recap)

Problem 27: Disjoint vs. Independent

(slide 7; transcribed from speaker notes)

Suppose $P(E) = 0.3$, $P(F) = 0.5$, and E and F are mutually exclusive. Are E and F independent?

Solution. No. Because they are disjoint, $P(EF) = 0$. But $P(E)P(F) = 0.3 \cdot 0.5 = 0.15 \neq 0$. In general, mutually exclusive events with positive probability are never independent: knowing that E happened tells you F did not. ■

Problem 28: Monty Hall

(slide 10)

There are 3 doors and 1 prize, equally likely to be behind any door. You pick door A. Monty, who knows where the prize is, opens a different door with no prize.

- (a) Should you switch?
- (b) In a 1000-door version, you pick 1 door and Monty opens 998 empty doors. Do you still want to stay?

Solution.

- (a) Condition on where the prize is. If you stay, you win only if the prize was behind A, so $P(\text{win by staying}) = \frac{1}{3}$. If you switch, you win whenever your first pick was wrong, because Monty must open the only other empty door, leaving the prize behind the door you switch to. So $P(\text{win by switching}) = P(\text{prize not behind A}) = \frac{2}{3}$. You should switch.
- (b) Staying wins only if your first pick was right: $\frac{1}{1000}$. Switching wins otherwise: $\frac{999}{1000}$.

Monty's choice depends on where the prize is, so it carries information. ■

7.2 Counting

Problem 29: Extra: Stars and Bars

(slide 17; wording reconstructed)

In how many ways can 10 identical units be distributed among 3 labeled bins?

Solution. Line up 10 stars and $3 - 1 = 2$ bars. Each arrangement corresponds to exactly one way to split the units (the stars before the first bar go in bin 1, and so on). So the number of ways is

$$\binom{10 + (3 - 1)}{2} = \binom{12}{2} = 66.$$

■

Problem 30: Bunnies and Foxes

(slide 21)

There are 4 bunnies and 3 foxes in a box. Draw 3. What is $P(1 \text{ bunny and } 2 \text{ foxes})$?

Solution. Make the animals distinct ($B_1, \dots, B_4$ and F_1, F_2, F_3) so that outcomes are equally likely.

Unordered: $|S| = \binom{7}{3} = 35$ and $|E| = \binom{4}{1}\binom{3}{2} = 12$, so $P = \frac{12}{35}$.

Ordered: $|S| = 7 \cdot 6 \cdot 5 = 210$. For E , choose which of the 3 positions holds the bunny (3 ways), then fill the slots in order ($4 \cdot 3 \cdot 2$ ways): $|E| = 3 \cdot 24 = 72$. So $P = \frac{72}{210} = \frac{12}{35}$.

Either way gives the same answer, as long as S and E are counted the same way. *Trap:* the “indistinct” sample space $\{3 \text{ bunnies}, 2B1F, 1B2F, 3 \text{ foxes}\}$ has 4 outcomes that are *not* equally likely. ■

7.3 The Binomial

Problem 31: Where $\binom{n}{k}$ Comes From

(slide 42)

Flip a coin with $P(\text{heads}) = p$ four times. What is $P(\text{exactly 2 heads})$? What is it for a fair coin?

Solution. The sequences with exactly 2 heads are HHTT, HTHT, HTTH, THHT, THTH, TTHH, so there are $\binom{4}{2} = 6$. Since the flips are independent, each sequence has probability $p^2(1-p)^2$. The sequences are mutually exclusive, so we add:

$$P(\text{exactly 2 heads}) = 6p^2(1-p)^2.$$

For a fair coin ($p = \frac{1}{2}$), this is $\frac{6}{16} = \frac{3}{8}$. ■

7.4 Moments

Problem 32: Lying with Statistics: Class Sizes

(slide 53)

A school has three classes, of sizes 5, 10, and 150. What is the expected class size if you pick a *class* at random? If you pick a *student* at random? Which number would a university brochure report?

Solution. Picking a class at random, each size has probability $\frac{1}{3}$: $E[\text{size}] = \frac{5+10+150}{3} = 55$.

Picking a student at random, a class of size s is chosen with probability $\frac{s}{165}$:

$$E[\text{size}] = \frac{5 \cdot 5 + 10 \cdot 10 + 150 \cdot 150}{165} = \frac{22,625}{165} = \frac{4525}{33} \approx 137.1.$$

The brochure would report 55, but a typical student experiences a class of about 137. ■

Problem 33: Matching Gifts

(slide 57)

n people each bring a gift, and the gifts are handed back in a random order. On average, how many people get their *own* gift back?

Solution. Computing $\sum_k k \cdot P(X = k)$ directly would need every $P(X = k)$, and even $P(X = 0)$ is hard. Instead, let $I_i = 1$ if person i gets their own gift. Person i 's gift is equally likely to go to any of the n people, so $E[I_i] = P(I_i = 1) = \frac{1}{n}$. Writing $X = I_1 + \cdots + I_n$, linearity gives

$$E[X] = \sum_{i=1}^n E[I_i] = n \cdot \frac{1}{n} = 1,$$

for every n . The indicators are dependent (if $n-1$ people match, so does the last), but linearity of expectation doesn't need independence. ■

Problem 34: Shaq at the Free-Throw Line

(slide 61)

Shaq made 53% of his free throws.

- (a) Let X be the points from one free throw. Find $E[X]$ and $\text{Var}(X)$.
- (b) Let Y be the points from 500 free throws. Find $E[Y]$, $\text{Var}(Y)$, and $\text{SD}(Y)$.
- (c) If Shaq were 10 percentage points better, how many more free throws would he expect to make out of 500?

Solution.

(a) $X \sim \text{Bern}(0.53)$, so $E[X] = 0.53$ and $\text{Var}(X) = 0.53 \cdot 0.47 = 0.2491$.

(b) Assuming independent throws, $Y \sim \text{Bin}(500, 0.53)$. So

$$E[Y] = 500 \cdot 0.53 = 265, \quad \text{Var}(Y) = 500 \cdot 0.53 \cdot 0.47 = 124.55, \quad \text{SD}(Y) \approx 11.2.$$

Typical totals land within about 11 points of 265.

(c) With $p = 0.63$, $E[Y] = 500 \cdot 0.63 = 315$, which is 50 more expected makes.

■